

RECON



Basic Reconnaissance Practical Report

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Overview

Reconnaissance (Recon) is the process of gathering information about a target before further analysis. In this practical work, I used five tools:

- WhatWeb
- Subfinder
- PureDNS
- HTTPX
- Katana

Target:

```
https://www.ewubd.edu
```

Step 1: Technology Discovery using WhatWeb

Objective

Identify the technologies, frameworks, web server, and libraries used by the EWU website.

Command

```
whatweb-v https://www.ewubd.edu
```

Information Found

Technology	Result
Web Server	Apache 2.4.34
Operating System	Ubuntu Linux
Frontend Framework	Bootstrap
JavaScript Library	jQuery 3.0.0, 3.4.0
Analytics	Google Analytics
Tag Manager	Google Tag Manager
Cookie	october_session
Security	HttpOnly Cookie
Emails	admissions@ewubd.edu, info@ewubd.edu

What I Learned

- EWU uses Apache on Ubuntu.
- Bootstrap and jQuery are used for the frontend.

- Session cookies are protected with `HttpOnly`.
- Public email addresses can be discovered.
- Technology discovery helps us understand the target before moving forward.

```
(kali@kali)~[~/WhatWeb-0.6.4]
$ whatweb -v https://www.ewubd.edu
WhatWeb report for https://www.ewubd.edu
Status      : 200 OK
Title       : East West University
IP          : <Unknown>
Country     : <Unknown>

Summary    : Apache[2.4.34], Bootstrap, Cookies[october_session], Email[admissions@ewubd.edu,info@ewubd.edu], Google-Analytics[Univers
al][UA-146902251-1], Google-Tag-Manager[UA-146902251-1], HTML5, HTTPServer[Ubuntu Linux][Apache/2.4.34 (Ubuntu)], HttpOnly[october_se
ssion], JQuery[3.0.0,3.4.0], Lightbox, Meta-Author[East West University], Open-Graph-Protocol[website], PasswordField[password], Scri
pt[text/javascript], X-UA-Compatible[IE=edge]

Detected Plugins:
[ Apache ]
The Apache HTTP Server Project is an effort to develop and
maintain an open-source HTTP server for modern operating
systems including UNIX and Windows NT. The goal of this
project is to provide a secure, efficient and extensible
server that provides HTTP services in sync with the current
HTTP standards.

Version      : 2.4.34 (from HTTP Server Header)
Google Dorks : (3)
Website     : http://httpd.apache.org/
```

Step 2: Subdomain Discovery using Subfinder

Objective

Find subdomains related to EWU.

Command

```
subfinder-d ewubd.edu > sub.txt
```

Result

Subfinder discovered **75 subdomains**.

```
(kali㉿kali)-[~/WhatWeb-0.6.4]
$ subfinder -d ewubd.edu > sub.txt

Subfinder
projectdiscovery.io

[INF] Current subfinder version v2.14.0 (latest)
[INF] Loading provider config from /home/kali/.config/subfinder/provider-config.yaml
[INF] Enumerating subdomains for ewubd.edu
[INF] Found 75 subdomains for ewubd.edu in 1 second 599 milliseconds
```

Interesting Subdomains

```
portal.ewubd.edu
admission.ewubd.edu
result.ewubd.edu
lib.ewubd.edu
etender.ewubd.edu
faportal.ewubd.edu
dspace.ewubd.edu
clubs.ewubd.edu
jobs.ewubd.edu
ccc.ewubd.edu
```

What I Learned

- One organization may have many subdomains.
- Different subdomains serve different purposes.
- Not every discovered subdomain is active.

```

(kali㉿kali)-[~/WhatWeb-0.6.4]
$ cat sub.txt
crt.ewubd.edu
devportal.ewubd.edu
vm03.ewubd.edu
vm04.ewubd.edu
lib-1tb.ewubd.edu
media.ewubd.edu
webmail.ewubd.edu
filetracker.ewubd.edu
alumnet.ewubd.edu
convo2021.ewubd.edu
etender.ewubd.edu
results.ewubd.edu
flass.ewubd.edu
mon01.ewubd.edu
v2.ewubd.edu
alumni.ewubd.edu
gsdl.ewubd.edu
new1.ewubd.edu
faportal.ewubd.edu
news.ewubd.edu

```

Step 3: Validate Subdomains using PureDNS

Objective

Verify which discovered subdomains are valid.

Command

```

puredns resolve sub.txt \
-r ~/.config/puredns/resolvers.txt \
-w valid.txt

```

Result

- Subdomains found by Subfinder: **75**

- Valid subdomains after DNS validation: **54**

Examples

```
portal.ewubd.edu
admission.ewubd.edu
result.ewubd.edu
fbe.ewubd.edu
fse.ewubd.edu
flass.ewubd.edu
```

What I Learned

- Not all discovered subdomains are valid.
- PureDNS removes invalid domains.
- The output becomes cleaner for the next tools.

```
(kali㉿kali)-[~/WhatWeb-0.6.4]
└─$ puredns resolve sub.txt \
-r ~/.config/puredns/resolvers.txt \
-w valid.txt
```



puredns v2.1.1

Fast and accurate DNS resolving and bruteforcing

Crafted with <3 by @d3mondev
<https://github.com/sponsors/d3mondev>

```
[+] Mode : resolve
[+] File : sub.txt
[+] Resolvers : /home/kali/.config/puredns/resolvers.txt
[+] Rate Limit : unlimited
[+] Rate Limit (Trusted) : 500 qps
[+] Wildcard Threads : 100
[+] Wildcard Tests : 3
[+] Write Domains : valid.txt
```

Resolving domains with public resolvers

```
[ETA 00:00:00] | ██████████| 75/75 rate: 75 qps (time: 00:00:00)
```

Detecting wildcard root subdomains

[illegible]

Validating domains against trusted resolvers

```
[ETA 00:00:00] | ██████████| 54/54 rate: 54 qps (time: 00:00:00)
```

Found 54 valid domains:

```
admission.ewubd.edu /img/themes/east-west-university/assets/default/plugins/devbridge_autocomplete/jquery.autocomplete.min.js
alumni.ewubd.edu /img/themes/east-west-university/assets/default/plugins/devbridge_autocomplete/jquery.autocomplete.min.js
lib.ewubd.edu /img/themes/east-west-university/assets/default/plugins/select2/select2.min.js
portal.ewubd.edu /img/themes/east-west-university/assets/default/js/initPBCalendar.js
www.ewubd.edu /img/themes/east-west-university/assets/default/js/initPBCalendar.js
fse.ewubd.edu /img/themes/east-west-university/assets/default/js/initPBCalendar.js
flass.ewubd.edu /img/themes/east-west-university/assets/default/js/initPBCalendar.js
dSPACE.ewubd.edu /img/themes/east-west-university/assets/default/js/initPBCalendar.js
etender.ewubd.edu /img/themes/east-west-university/assets/default/js/initPBCalendar.js
filetracker.ewubd.edu /img/themes/east-west-university/assets/default/js/initPBCalendar.js
clubs.ewubd.edu /img/themes/east-west-university/assets/default/js/initPBCalendar.js
fbe.ewubd.edu /img/themes/east-west-university/assets/default/js/initPBCalendar.js
faportal.ewubd.edu /img/themes/east-west-university/assets/default/js/initPBCalendar.js
modules.ewubd.edu /img/themes/east-west-university/assets/default/js/initPBCalendar.js
ccc-server.ewubd.edu /img/themes/east-west-university/assets/default/js/initPBCalendar.js
gsdl.ewubd.edu /img/themes/east-west-university/assets/default/js/initPBCalendar.js
```

```

(kali㉿kali)-[~/WhatWeb-0.6.4]
$ httpx -l valid.txt -sc -title
https://cdnjs.cloudflare.com/ajax/libs/moment.js/2.22.1/moment.min.js
https://www.ewubd.edu/home/east-west-university/assets/default/js/pb.calendar.js
https://www.ewubd.edu/home/east-west-university/assets/default/js/ui/effect.min.js
https://www.ewubd.edu/home/east-west-university/assets/default/js/pluginsUpdated.min.js
https://www.ewubd.edu/home/east-west-university/assets/default/plugins/lightbox.js
https://cse.google.com/cse.js?cx=017461950370971895329:0tyupils2d8
https://www.ewubd.edu/home/east-west-university/assets/js/framework.js
https://www.ewubd.edu/home/east-west-university/assets/js/framework.extras.js
projectdiscovery.io
[INF] Current httpx version v1.10.0 (latest)
[WRN] UI Dashboard is disabled, Use -dashboard option to enable
https://ccc.ewubd.edu [200] [A{{ title }}]
https://jobs.ewubd.edu [200]
https://alumni.ewubd.edu [200] [A{{ title }}]
https://portal.ewubd.edu [200] [EWU Student Panel: Login]
https://filetracker.ewubd.edu [200] [EWU File Tracking]
https://etender.ewubd.edu [200] [EWU eTender]
https://vm01.ewubd.edu [200]
https://modules.ewubd.edu [200] [EWU Modules]
https://admission.ewubd.edu [200] [East West University - Online Admission Form]
https://dSPACE.ewubd.edu [200] [Apache2 Debian Default Page: It works]
https://ewjbss.ewubd.edu [302]
https://ewjh.ewubd.edu [302]
https://vm03.ewubd.edu [200]

```

```

https://admission.ewubd.edu [200] [East West University - Online Admission Form]
https://dSPACE.ewubd.edu [200] [Apache2 Debian Default Page: It works]
https://ewjbss.ewubd.edu [302]
https://ewjh.ewubd.edu [302]
https://vm03.ewubd.edu [200]
https://home.ewubd.edu [200] [East West University - Online Admission Form]
https://ccc-server.ewubd.edu [404]
https://faportal.ewubd.edu [200] [Educational Resource Management]
https://vm04.ewubd.edu [200]
https://crtproposals.ewubd.edu [302]
https://alumnet.ewubd.edu [302]
https://vm02.ewubd.edu [200]
https://crt.ewubd.edu [200] [EWUCRT | Center for Research and Training (CRT), EWU]
https://alumniportal.ewubd.edu [200] [EWU Alumni Portal - East West University]
https://jobs1.ewubd.edu [200] [A{{ title }}]
https://vm05.ewubd.edu [200]
https://fbe.ewubd.edu [200]
https://fse.ewubd.edu [200]
https://tracking.ib88647.notification.ewubd.edu [503]
https://flass.ewubd.edu [200]
https://csefest.ewubd.edu [200] [Index of /]
https://iikm.ewubd.edu [302] [302 Found]
https://result.ewubd.edu [200] [East West University Result]
https://clubs.ewubd.edu [200] [East West University]
https://www.ewubd.edu [200] [East West University]
http://lib.ewubd.edu [200] [Dr. S. R. Lasker Library | Gateway to knowledge]
https://edc-seip.ewubd.edu [200] [EDC-SEIP - Skills for Employment Investment Program]
http://gsdl.ewubd.edu [200] [Dr. S. R. Lasker Library catalog]
http://opac.ewubd.edu [200] [Dr. S. R. Lasker Library catalog]

```

Interesting Findings

Student Portal

```
https://portal.ewubd.edu
```

Used by students for academic activities.

File Tracker

```
https://filetracker.ewubd.edu
```

Dedicated file tracking system.

DSpace

```
https://dspace.ewubd.edu
```

Returned:

```
Apache2 Debian Default Page: It works
```

This means the default Apache page is still accessible.

Library Services

```
http://lib.ewubd.edu  
http://opac.ewubd.edu  
http://gsdl.ewubd.edu
```

Provide library access.

What I Learned

- HTTPX identifies active websites.
- HTTP status codes provide useful information.
- Page titles reveal the purpose of each subdomain.

- Some systems are unavailable or misconfigured.

Step 5: URL Discovery using Katana

Objective

Discover URLs, pages, JavaScript files, and hidden resources.

Command (Standard Crawling)

katana-u <https://www.ewubd.edu>

Command (Headless + JavaScript)

katana-u <https://www.ewubd.edu-headless-jc>

```
(kali@kali)~[WhatWeb-0.6.4]
$ katana -u https://www.ewubd.edu

[INF] Current katana version v1.6.1 (latest)
[INF] Started standard crawling for ⇒ https://www.ewubd.edu
https://www.ewubd.edu
https://www.ewubd.edu/themes/east-west-university/assets/default/js/initPBCalender.js
https://www.ewubd.edu/themes/east-west-university/assets/default/js/pluginsUpdated.min.js
https://www.ewubd.edu/modules/system/assets/js/framework.extras.js
https://www.ewubd.edu/modules/system/assets/js/framework.js
https://www.ewubd.edu/themes/east-west-university/assets/default/plugins/lightbox/js/lightbox.min.js
https://www.ewubd.edu/themes/east-west-university/assets/default/plugins/select2/js/select2.min.js
https://www.ewubd.edu/themes/east-west-university/assets/default/plugins/devbridge/jquery.autocomplete/jquery.autocomplete.min.js
https://www.ewubd.edu/themes/east-west-university/assets/default/js/main.js
https://www.ewubd.edu/themes/east-west-university/assets/default/js/ui/effect.min.js
https://www.ewubd.edu/themes/east-west-university/assets/default/js/pb.calendar.js
https://www.ewubd.edu/themes/east-west-university/assets/default/js/magnific-popup.min.js
https://www.ewubd.edu/themes/east-west-university/assets/default/js/owl.carousel.min.js
https://www.ewubd.edu/themes/east-west-university/assets/default/js/thumbnaill-slider.js
https://www.ewubd.edu/themes/east-west-university/assets/default/js/aos.js
https://www.ewubd.edu/themes/east-west-university/assets/default/js/slick.min.js
```

```

Session Actions Edit View Help
(kali㉿kali)-[~]
$ katana -u https://www.ewubd.edu -headless -jc
[INF] Current katana version v1.6.1 (latest)
https://dbaconf.ewubd.edu/
https://crt.ewubd.edu/notice-detail/136th-research-seminar-ewucrt
https://result.ewubd.edu
https://crt.ewubd.edu/notice-detail/2nd-call-papers-east-west-journal-humanities-vol13-2026
https://crt.ewubd.edu/notice-detail/call-research-proposals-round-19
https://crt.ewubd.edu/notice-detail/2nd-call-abstracts-published-research-papers-vol-20-2025
https://result.ewubd.edu/
https://www.ewubd.edu
https://fse.ewubd.edu
https://flass.ewubd.edu
https://fbe.ewubd.edu
https://fse.ewubd.edu/computer-science-engineering
https://fse.ewubd.edu/electrical-electronic-engineering
https://fse.ewubd.edu/genetic-engineering-biotechnology
https://fse.ewubd.edu/pharmacy-department
https://fse.ewubd.edu/civil-engineering
https://fse.ewubd.edu/mathematical-physical-science

```

URLs Found

Important Systems

<https://portal.ewubd.edu>
<https://result.ewubd.edu>
<https://admission.ewubd.edu>
<https://ccc.ewubd.edu>
<https://alumni.ewubd.edu>

Administrative Pages

<https://www.ewubd.edu/admin-office/office-of-the-registrar>
<https://www.ewubd.edu/admin-office/office-of-the-controller-of-examination>
<https://www.ewubd.edu/admin-office/office-of-the-proctor>

Notice and Event Pages

<https://www.ewubd.edu/notice-board>
<https://www.ewubd.edu/news>

```
https://www.ewubd.edu/events
```

JavaScript Files

```
https://www.ewubd.edu/themes/east-west-university/assets/default/js/main.js
https://www.ewubd.edu/modules/system/assets/js/framework.js
https://www.ewubd.edu/themes/east-west-university/assets/default/js/pluginsUpdated.min.js
```

API Endpoint Found

```
https://www.ewubd.edu/api/events
```

Suspicious/Interesting URLs

```
https://www.ewubd.edu/backend/backend/auth/restore
```

This URL appears related to backend authentication and password restoration.

What I Learned

- Katana discovers URLs automatically.
- Headless mode finds more content than standard mode.
- JavaScript files may contain APIs and hidden endpoints.
- APIs, login systems, and backend pages are important for further analysis.
- Large websites contain hundreds of URLs, so filtering is necessary.

Overall Workflow

```
WhatWeb
  ↓
Subfinder
  ↓
PureDNS
```

↓
HTTPX
↓
Katana

Purpose of Each Tool

Tool	Purpose
WhatWeb	Identify technologies
Subfinder	Find subdomains
PureDNS	Validate subdomains
HTTPX	Find active websites
Katana	Discover URLs and resources